

Problem A:

LLM-Assisted Netlist Exploration and Transformation

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Q&A

Q1. The examples in the document use named instances such as "and g1(w1, a, b);". In standard Verilog, unnamed instances are also valid, e.g., "and (w1, a, b);". Will all gate instances in the test benchmarks always have explicit instance names, or should we also handle unnamed instances?

A1. Yes, the test benchmarks will always have explicit instance names.

Q2. Section 3.2.a specifies 8 primitive gates and dff. Can we assume that the netlists will only contain Verilog built-in primitive keywords (and, or, nand, nor, not, buf, xor, xnor, dff), and not technology library cell names (e.g., NAND2X1, AOI22X1)?

A2. Yes, the netlists will only contain the built-in primitive keywords.

Q3. Will any sample test cases (input netlists + natural language request sequences) be released before the alpha submission deadline?

A3. Yes, we will release the sample test cases before the end of April.

Q4. I wonder which exact LLM models will be provided during the alpha, beta, and final testing phases. Could you please specify them?

Additionally, I would like to know if participants are permitted to fine-tune these models, or if the models can only be utilized via API as an agent or similar framework.

A4. ChatGPT 4o mini and Claude 4.5 Haiku will be provided. These models can only be used via API.

Q5. While reviewing the problem statement, I have encountered several technical questions regarding the evaluation environment and testcase specifications. I would greatly appreciate your clarification on the following points:

1. Executable Format

The problem statement specifies the executable naming convention (e.g., cada0001_alpha) but does not restrict the implementation language or packaging method. Is it acceptable to submit a self-contained executable packaged with tools such as PyInstaller, or must the submission be a natively compiled binary?

2. Network Access During Evaluation

Since the system is required to call an external LLM service (e.g., OpenAI or Anthropic API) during evaluation, will the TSRI Machine have outbound internet access to reach these cloud endpoints? This is critical to our architecture design.

3. Working Directory and File Path Convention

When the executable is invoked, what is the assumed working directory? Should all input Verilog files be expected relative to the working directory, and should output files (e.g., result netlists and log files) also be written relative to it?

4. DFF Module Definition in Input Netlists

Will the DFF module definition (as shown in Figure 2 of the problem statement) always be included in the input Verilog file, or should we assume the exact interface from Figure 2 and treat DFF instances as black boxes during parsing?

5. Availability of Sample Testcases

Are there any sample or example testcases available for download? Having concrete examples would help me better understand the expected input/output format and the range of task complexity. I would also like to ask whether additional testcases will be released before the final evaluation.

6. Validity Guarantee of Input Verilog Netlists

Will the input Verilog netlists provided during evaluation always be syntactically and semantically valid (i.e., free of undefined signals, unconnected ports, or non-standard

primitives)? Knowing whether defensive error handling for malformed inputs is necessary would significantly affect my parser design.

A5.

~~1. Submission of a PyInstaller package in a Docker image is acceptable.~~

1. It is acceptable only if the program can be executed on the TSRI machine. Please note that, during evaluation, internet access is allowed only for model APIs.

2. Internet access is available only for the designated model API endpoints during evaluation; general internet access is not provided.

~~3. Yes. The Docker working directory is /app. Input files will be mounted under /app, and your program should read the design from the path given in the prompt and write all output files to the same testcase directory.~~

3. Yes, all input and output files should be relative to the working directory.

4. The DFF will appear as a primitive gate in the input Verilog file.

5. The test cases have been uploaded.

6. There may be floating or unconnected ports in the Verilog netlist.

Q6. We are a participating team for Problem A: LLM-Assisted Netlist Exploration and Transformation. As we are currently designing our agentic framework, we would like to seek clarification on several technical aspects to ensure our implementation is compatible with the official evaluation environment:

1. Compatibility with Model Context Protocol (MCP):

To enhance tool-use stability across different LLMs, we are considering using the Model Context Protocol (MCP). However, we have concerns regarding the Docker evaluation environment:

a. Does the environment allow Inter-Process Communication (IPC) via stdio or local pipes between the Agent host and a tool server?

b. Will the environment have internet access to install additional packages (e.g., the mcp Python SDK), or should we assume a strictly offline/pre-installed environment?

c. If IPC is restricted or the environment is offline, an MCP-based architecture might fail. Could you provide guidance on whether MCP is supported or if we should strictly use standard Function Calling?

2. Scope of Required Functionalities:

The Problem A description provides several examples of tool functions. We would like to confirm whether the examples provided in the document constitute the

exhaustive list of tools/APIs we need to implement, or if there will be additional, unlisted functions introduced during the final evaluation?

3. Evaluation Model Specifications:

Figure 6 of the configuration file mentions gpt-4o-mini and claude-haiku-4-5. Is it confirmed that only these two specific models will be used for the final evaluation? Or is there a possibility that other models (e.g., larger-scale models or different versions) will be utilized as well?

A6.

~~1a. IPC is acceptable, but the MCP server must reside on the local machine.~~

1a. Docker submissions are not supported. IPC is acceptable, but the MCP server must reside on the local machine.

~~1b. Internet access is allowed only for model APIs; contestants should assume no general internet access for package installation.~~

1b. Docker submissions are not supported. Internet access is allowed only for model APIs; contestants should assume no general internet access for package installation.

1c. MCP is supported, but the environment is offline.

2. There will be hidden prompts in the final evaluation. Contestants should not only consider the tool functions in the document.

3. Only these two models will be used for evaluation. We will switch to different models only if both of them are deprecated.

Q7. Will the contest provide us the api keys for openai and anthropic? Or, do we have to use our own ai keys?

A7. Contestants should use their own API keys during development.

Q8. In Problem A, the “cost” and “cost_min” are mentioned, cost_min/cost decides our rank. What is “cost” exactly? It isn’t stated in the pdf.

A8. We will define the cost inside the prompt.

Q9. In Problem A, for optimization testcases, solutions are ranked by a cost ratio relative to cost_best.

Is the cost simply the objective stated in the request (e.g., minimize gate count), or is it computed differently?

A9. The cost will be defined inside the prompt.

Q10. EDA Engine: Must the EDA back-end be built entirely from scratch, or may we use existing tools (commercial or open-source) as the implementation — e.g., Cadence Genus/JasperGold via TCL scripting, or open-source tools such as Yosys or Icarus Verilog?

A10. We have no restrictions on EDA back-end tools.

Q11. LLM Evaluation: Will the final evaluation be run using one of the two specified LLMs (GPT-4o mini, Claude Haiku 4.5) or both? If both, how are the scores combined?

A11. We will consider each LLM as a separate case.

Q12. I have a few questions regarding the competition setup and APIs:

In the problem description, the dff module definition seems inconsistent with the dff used in the provided gate-level Verilog test examples. For example, in test35.v:

```
dff g10537(.RN(n1), .SN(1'b1), .CK(n0), .D(n49[2]), .Q(n194[2]));
```

However, the problem statement defines the dff module as:

```
module dff (  
    input wire clk,      // Clock input  
    input wire rst_n,    // Asynchronous active-low reset  
    input wire d,        // Data input  
    output reg q         // Data output  
);
```

The number of the ports do not appear to match.

A12. The dff provided in the released test case is the correct one. .SN is the active-low set signal.

Q13. Is the provided API fully compatible with the OpenAI API format and calling conventions? For example, does it support features such as Function Calling / Tool Calling in the same way as the OpenAI API?

A13. Yes, the provided OpenAI API key is fully compatible with the OpenAI API format. During development, contestants should use their own API keys. The contest will replace the API key with its own during evaluation.

Q14. Since LLM responses are inherently non-deterministic, the same question may produce different answers with different wording while still having the same meaning. Therefore, We would like to clarify:

1. Will the evaluation be based on semantic equivalence rather than exact string matching?
2. Will the organizer use deterministic settings such as fixed random seeds or temperature = 0 during evaluation?
3. If an answer is semantically correct but phrased differently from the reference answer, will it still receive full credit?
4. Is there any public information about the evaluation metric or judging method (e.g., keyword matching, embedding similarity, LLM-as-a-judge, etc.)?

We would like to better understand the evaluation assumptions to ensure our implementation aligns with the official judging process.

A14.

1. Yes.
2. No.
3. Yes
4. The evaluation uses LLM-as-a-judge methodology based on semantic equivalence.

Q15. Are teams permitted to bundle statically-built third-party binaries (e.g., a standalone ABC build) inside the submission package and invoke them as subprocesses from the team executable? If yes, are there constraints on binary size, source-availability requirements, or license (must it be open-source / a permissive license)?

A15. Yes, third-party binaries are permitted with no size limit.

Q16. For "list all X" prompts where the answer can contain tens of thousands of items (e.g., test35 contains 19,682 NAND gates), is the grading pipeline expected to consume the full enumerated list, or is there a documented response-length / character cap above which the output is treated as malformed? If a cap exists, could you share the exact threshold so we can size our internal output behavior accordingly?

A16. Yes, full enumerated lists are expected. For large result sets, write the complete list to a file and provide the file path in your response.

Q17. What are the per-testcase (and per-run) RAM and disk limits in the evaluation environment? This determines whether we should default to a SAT-based equivalence checker vs. a BDD-based one, and how aggressively we may cache intermediate Boolean structures or AIG dumps.

A17. Please refer to the TSRI machine specifications.

Q18. For the GPT-4o-mini and Claude 4.5 Haiku endpoints provided via the evaluation config, are there per-request rate limits, per-testcase token budgets, or hard request timeouts we should design the agent to respect? Knowing the official policy will help us tune retry / back-off and fallback to rule-based routing safely.

A18. Rate limits follow the policies of GPT-4o-mini and Claude 4.5 Haiku. There is no token limit. The runtime limit is 60 seconds for each basic operation request (as defined in Section 4.1) and 300 seconds for all other requests.

Q19. Will an official cost estimator or evaluation tool be provided so that participants can locally validate their results and estimate their scores before submission?

A19. No

Q20. For the Final Submission evaluation, will the scoring be based on both public testcases and hidden testcases? If so, could you clarify how they will contribute to the final score?

A20. The final evaluation will include hidden test cases.

Q21. During implementation we have encountered several specification points where the intended interpretation is not obvious from the problem statement and the public test cases. We would be very grateful for your clarification, so that our system can align with the official judging criteria.

1. Definition of "constant" inputs / signals

- Relevant questions:
 - testcase36 Q7 — "Report any AND gates with a constant 0 input in this design."
 - testcase32 Q17 — "Simplify the reported NAND gates by propagating their constant inputs."
 - testcase31 Q8 — "Is output n16 always 0 regardless of all inputs?"
 - Does "constant" refer only to pins structurally tied to 1'b0 /1'b1 in the netlist, or does it also include signals that are functionally constant (provably 0/1 through upstream logic, e.g., via constant propagation or SAT)?
 - For testcase31 Q8 specifically: how should DFF initial state and unknown (X) values be treated when judging "always 0"?

2. Fan-in depth across sequential elements

- Relevant question pattern: "Which output bit has the deepest fan-in logic cone?"
- Should the fan-in cone terminate at DFF outputs (treating DFF.Q as a primary input, i.e., combinational depth only), or should it traverse through DFFs into prior cycles (sequential depth)?
- If traversal is expected, how should feedback loops and initial state be handled?

3. Path enumeration output size

- Relevant question pattern: "Provide a complete enumeration of paths between n0[1] and n63[0]." (For testcase14 this yields 289,366 paths; a literal listing exceeds several hundred MB.)

- Is "complete enumeration" expected to list every path literally, or is a path count plus representative samples acceptable?

- Is there a maximum response size or maximum number of paths the judge will accept? How is the boundary drawn between count-style questions and enumerate-style questions?

4. Judging criteria for transformation / rewrite questions

- Relevant questions:

- testcase32 Q7 — "Try to replace all 2-input NAND gates that have one input tied to constant 1 with inverters. Ensure the design functionality does not change."

- testcase32 Q17 — NAND constant propagation.

- Is functional equivalence the sole judging metric, or are secondary metrics (gate count, area, logic depth, fan-out, etc.) also scored?

- When multiple legal rewrites are possible (e.g., partial vs. full replacement, or different gate substitutions), which output is preferred?

5. Concurrent LLM API calls

- Are concurrent / asynchronous LLM API calls permitted during evaluation, i.e., may we issue multiple LLM requests in parallel for a single test case (or across test cases)?

- If yes, are there rate limits, per-question time budgets, or per-test total time budgets we should design around?

6. Multithreaded program design

- Independently of LLM calls, is our program itself allowed to be multithreaded / multi-process during evaluation (e.g., parallel netlist analysis, parallel SAT queries, worker pools)?

- Are there constraints on CPU cores, memory, or process count enforced by the judging environment?

A21.

1. A signal is functionally constant if it provably remains 0 or 1 for all possible inputs. The DFF initial state is 0, and X values are ignored.

2. Combinational depth only; treat DFF.Q outputs as primary inputs.

3. Complete enumeration means list every path literally. For very large result sets, write the output to file and provide the file path in the response.
4. Functional equivalence is the primary judging metric. Secondary metrics will be explicitly specified in the prompt if they are used for scoring. When multiple legal rewrites are possible, the prompt will clarify which optimization criteria to prioritize.
5. Only single-threaded execution is permitted, with one request at a time. Rate limits follow the policies of GPT-4o-mini and Claude 4.5 Haiku. There is no token limit. The runtime limit is 60 seconds for each basic operation request (as defined in Section 4.1) and 300 seconds for all other requests.
6. Only single-threaded execution is permitted. Please refer to the TSRI machine specifications for hardware constraints.

Q22. Regarding the 40 provided testcases, do the Verilog features appearing in these testcases already cover all possible Verilog constructs that may appear in the final evaluation benchmarks?

If there are additional Verilog features or syntax elements that do not appear in the released 40 testcases but may still appear during evaluation, could you please provide a list of those possible features?

A22. The Verilog syntax will follow the rules defined in Section 3.2 of the problem description.

Q23. How exactly is the cost calculated/evaluated during the contest?

A23. We will define the cost in the prompt.

Q24. Will participants be provided with either GPT-4o Mini or Claude Haiku 4.5 API credits?

A24. No, contestants should use their own API during development.

Q25. I have a question about the expected judging criterion for test01. From the prompt, my understanding is that this testcase mainly checks whether the submitted

flow can correctly load a Verilog netlist and write the current design back to an output Verilog file.

In my implementation, the design is loaded and written through OpenROAD. During this process, OpenROAD may canonicalize or reformat the Verilog output. For example, it may:

- expand grouped port declarations into separate declarations,
- reorder or split wire declarations,
- change formatting and indentation,
- emit ports and internal wires in a different textual order,
- preserve the same gates and connectivity while producing a textually

different Verilog file.

As a result, the output Verilog is not byte-for-byte identical to the input Verilog.

However, I checked that the design is logically equivalent and structurally consistent with the input. For example, the module name, primary inputs/outputs, number of instances, and gate-type counts are preserved, and the final equivalence check passes.

Could you please clarify the expected requirement for this testcase?

Specifically, for test01, is it sufficient that the written Verilog is functionally equivalent to the loaded design, or must the output file be textually identical to the input file?

A25. A textually different but functionally equivalent output is acceptable.

Q26. Are there answers (.v and prompt responses) for the 40 test questions?

For example:

test03: How many gates are in the fanin cone of primary output n15?

test06: Determine whether a combinational path from n0[0] to n4 exists that does not traverse node n719.

Those types of analysis are problematic.

A26. No, we won't provide the answers.

Q27. Do the 40 test questions count towards the score?

A27. The final evaluation will contain hidden test cases.

Q28. We previously inquired about the definition of "cost" in the scoring criteria for Problem A, and were told that it would be explained in the prompt. However, we have been unable to find any relevant explanation. Could you please provide more details — specifically, which prompt contains this information?

A28. Please refer to the [updated test cases](#).

Q29. For fanout-related queries and fanout optimization constraints, such as “maximum fanout 4,” should the fanout load count include all types of sink pins and output connections?

Specifically, should the following be counted as fanout loads?

- * Primitive gate input pins
- * DFF D pins
- * DFF clock pins
- * DFF reset/set pins, such as RN and SN
- * Primary output connections

For example, if a clock or reset signal drives many DFF clock/reset pins, should those DFF pins be counted as fanout loads for maximum-fanout constraints?

A29. Yes, those signals are considered fanout loads. If we only optimize a specific signal (e.g., reset), we will specify it in the prompt.

Q30. When prompts ask for functional equivalence, such as equivalence between internal signals, equivalence between the current design and the originally loaded netlist, or equivalence between pre- and post-transformation netlists, is combinational equivalence sufficient if DFFs are treated as sequential boundaries?

In other words, for designs containing DFFs, may we compare the combinational behavior around DFF boundaries, such as PI-to-PO, PI-to-DFF-D, DFF-Q-to-PO, and DFF-Q-to-DFF-D logic, rather than performing multi-cycle sequential equivalence checking?

A30. Yes, we only consider combinational equivalence.

Q31. We understand from the Q&A that final evaluation will include hidden prompts and that contestants should not only consider the example tool functions in the problem statement.

Could you clarify whether hidden prompts are still expected to remain within the same broad task families shown in the problem statement and released testcase prompts, such as:

- * Basic design loading and writing
- * Structural reporting
- * Path, fanin, fanout, and cone analysis
- * Boolean or equivalence analysis
- * Netlist transformation and optimization under explicitly stated constraints
- * Sequential structural reporting involving DFFs

Or may hidden prompts introduce fundamentally different classes of EDA tasks beyond these categories, such as timing analysis with annotated delays, placement/routing constraints, arithmetic datapath recognition, or protocol/state-machine analysis?

A31. We won't have timing or placement-related prompts in the hidden test cases.

Q32. We would like to ask about the requirements for analysis-related questions. In addition to outputting the results, are we also required to provide the corresponding cases? For example, in the case shown in the figure, it is difficult to determine from the question itself that path information should be included in the output.

Since the definition of the required output is a bit unclear to us, we are writing to ask for clarification. We would greatly appreciate your reply.

What is the maximum logic depth from input in0 to output out3?	<pre>The maximum logic depth from input in0 to output out3 is 5. One example of a longest combinational path (5 gate levels) is: in0 └─[NOT] U1: n_in0_n = ~in0 └─[AND] U5: n1 = n_in0_n & n_aux0 └─[XOR] U12: n2 = n1 ^ n_aux1 └─[NOR] U18: n3 = ~(n2 n_aux2) └─[BUF] U20: out3 = n3</pre>
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A32. No, you don't have to provide an example. If we need one, we will specify that in the prompt.

Q33. Regarding the question of adding one to depth, does an empty net need an extra node?

A33. No, you do not need to count an empty net as an extra depth node.

Q34. Some testcases may contain multiple transformation requests. For example:

- "Insert buffers wherever needed so that no gate drives more than 4 loads. Make sure nothing changes functionally."
- "Reconstruct the entire netlist using only AND and NOT gates while preserving functional equivalence."

In such cases, a later transformation may substantially modify or even overwrite the circuit structure produced by an earlier transformation. Furthermore, subsequent questions may ask for circuit properties such as gate count, depth, or fanout. Since multiple valid implementations may exist and different transformation sequences can lead to different results, it is unclear what the expected answer should be. Could you clarify how these cases will be evaluated?

A34. We have updated the test cases.

Q35. Could the contest provide explicit reference answers or expected outputs for the released testcases? Having clear reference results would help participants verify the correctness of their analysis and transformation tools.

A35. No, we won't provide reference answers.

Note:

Kindly note that the topic chairs have revised Q5 and Q6. The specific changes are marked in red for your convenience. Thank you.

Q36. I would like to ask, in Problem A (LLM-Assisted Netlist Exploration and Transformation) for alpha test submission should our system run only basic operation or is it required to run and correctly answer the analysis and transformation task (present in some of the test cases provided) also.

A36. Both approaches are acceptable for the alpha test.

Q37. We would like to clarify which API host should be used for the LLM calls in Problem A.

The released documents mention that the supported models are gpt-4o-mini and claude-haiku-4-5, and that the API is OpenAI-format compatible. However, we could not find the exact API host/domain that should be used.

Could you please confirm the correct host for each provider?

OpenAI host: ?

Anthropic host: ?

For example, should we use the public hosts such as:

`https://api.openai.com`

`https://api.anthropic.com`

or will the contest environment provide different hosts?

We also observed that the contest VM currently cannot resolve public API domains such as `api.openai.com`, so we would like to make sure our implementation uses the expected host in the evaluation environment.

A37. Yes, the hosts you provided are correct.

Q38. Could you please confirm whether we only need to upload the code files, without the need to run the code ourselves? In other words, will the evaluation team handle the execution and testing on their end?

A38. Yes, during the evaluation process, we will execute your program ourselves.

Q39. According to the Competition Host User Manual, the TSRI VM environment is based on RedHat 8 and provides information such as GCC, glibc, and Python

versions. However, in Q&A Q5-A5.3, it is stated that the Docker working directory is /app, with input files mounted under /app.

Therefore, we would like to ask whether our executable will be run directly on the TSRI VM / host machine, or inside a Docker container during the official evaluation? If it will be executed inside a Docker container, could you please provide the basic container environment information, such as the base image / OS and Python version?

A39. We will not support Docker image submissions. Please refer to the updated answer for Q&A Q5.

Q40. In the previous Q&A document, you mentioned that Docker image submissions are accepted. If we submit via Docker, what format is required for the Docker image? Would it be acceptable to place a .tar file (e.g., an image exported with "docker save") inside the alpha_test_submission folder? We would also like to confirm the expected executable / entrypoint name.

A40. Docker image submissions are not accepted. Please refer to the updated answer for Q&A Q5.

Q41. We would also like to reconfirm: during evaluation, does the contest provide the configuration file? If so, does its content match the example given in Section 6.2 of the problem description? In particular, is the official API key written into the configuration file (the nested api_key field in Section 6.2), or provided via an environment variable?

A41. During the evaluation, we will replace the API key in the configuration file with our own API key.

Q42. For analysis-type questions that have a clear conclusion (e.g., determining whether two nets are functionally equivalent), does the level of detail in the natural-language response affect scoring?

For example:

- Response A: "Yes."
- Response B: "Yes. The two nets are functionally equivalent because ...

(with supporting analysis such as paths, gates, or logic reasoning)."

Will both responses receive the same score if the conclusion is correct, or is supporting information required to obtain full credit?

A42. Yes, both answers are correct.

Q43. For tasks that require modifying a circuit and generating a new .v file, we would like to clarify the expected contents of the system log (<case_name>.log, specifically the #RESPONSE section).

Is it sufficient for the system to provide a natural-language summary of the modifications performed (similar to Figure 5 in the official documentation, where only the affected gates are described)?

Or is the LLM expected to output the complete modified Verilog netlist in the dialogue log as well?

In other words, can the full circuit modifications be provided exclusively through the generated output .v file, while the log only contains a summary of the changes?

A43. Yes, it is acceptable for the log to contain only a summary of the changes.

Q44. We would like to clarify how hard requirements are evaluated relative to functional equivalence.

In a previous Q&A, it was mentioned that functional equivalence is the primary evaluation criterion. Consider the following instruction:

"Replace the XOR gate using NOR and NOT gates."

Suppose our implementation replaces the XOR gate using only NOR gates while still preserving complete functional equivalence with the original circuit.

Would this solution be considered correct because the resulting circuit is functionally equivalent?

Or must the implementation strictly satisfy the literal instruction and explicitly use both NOR and NOT gates? If NOT gates are not used, would the testcase be considered incorrect due to violating a hard requirement?

More generally, which criterion takes precedence when there is a conflict between functional equivalence and the explicit gate-type requirements stated in the natural-language instruction?

A44. In this case, not using NOT gates is acceptable. If we strictly require both NOR and NOT gates, we will mention that in the prompt.

Q45. Will the scores obtained during the Alpha Test and Beta Test phases contribute to the final contest ranking, or are they provided solely for testing and feedback purposes?

A45. No. The Alpha Test and Beta Test scores are for testing purposes only. The final evaluation will use its own scores for the final ranking.

Q46. Multiplexer and DFF Enable/Hold Recognition Since MUX is not included in the supported primitive gate list, should a multiplexer be recognized only through a specific structural pattern, or through Boolean functionality regardless of its gate-level implementation?

For example: $D = (EN \& DATA) \mid (!EN \& Q)$ This is a conventional enable/hold multiplexer. When $EN=1$, the DFF loads DATA; when $EN=0$, it holds its current value through Q feedback. Should the following implementations all be recognized as the same enable/hold structure?

- The canonical AND/OR/NOT implementation above.
- A functionally equivalent NAND/NOR implementation.
- An implementation transformed using De Morgan's laws.
- An active-low implementation such as: $D = (!EN_N \& DATA) \mid (EN_N \& Q)$

In other words, should detection be based on:

- A. Specific structural gate patterns; or
- B. Boolean equivalence of the D-input function?

A46. B

Q47. Meaning of “Enable or Hold Structures Implemented Through AND Gates”

The prompt mentions enable or hold structures implemented through “multiplexers or AND gates.” Could you clarify what Boolean pattern should be considered an AND-gate enable/hold structure?

For example: $D = EN \& DATA$ When $EN=0$, this function forces $D=0$; it does not preserve the previous value of Q .

Should this still be reported as an enable or hold structure?

Alternatively, must a hold structure include feedback from the same DFF's Q output, such as: $D = (EN \& DATA) \mid (!EN \& Q)$

Please clarify which of these examples should be counted.

A47. We only consider this structure: $D = (EN \& DATA) \mid (!EN \& Q)$

Q48. Definition of Input Symmetry Should input symmetry include only positive permutation symmetry, where exchanging two inputs leaves the Boolean function unchanged?

For example: $f(a, b) = a \& b$ Swapping a and b does not change the function, so they are positively symmetric. Should complemented or negative symmetry also be reported?

For example: $f(a, b) = a \& !b$ This function is unchanged under the transformation: $(a, b) \rightarrow (!b, !a)$

Should a and b be considered symmetric in this case?

A48. Consider the positively symmetric only.

Q49. If two inputs do not appear in the functional support of an output, should they be considered vacuously symmetric?

A49. Yes.

Q50. May DFF.Q pseudo-primary inputs participate in symmetry analysis, or should only actual primary inputs be considered?

A50. Yes, DFF.Q signals are treated as pseudo-PIs and can be part of symmetry queries.

Q51. Definition of Cut Signals and Articulation Points Should cut-signal analysis operate on the directed combinational netlist graph or on an undirected graph?

For example: PI a -> gate1 -> net n1 -> gate2 -> PO y Removing n1 disconnects a from y, so n1 appears to be a cut signal.

However, consider a design with two independent paths: PI a -> n1 -> PO y1 PI b -> n2 -> PO y2 If removing n1 disconnects only the a-to-y1 path while the b-to-y2 path remains connected, should n1 still be reported as a cut signal?

In other words, does a cut signal mean:

- A. Removing it disconnects at least one previously connected PI-to-PO pair; or
- B. Removing it must disconnect all PI-to-PO connectivity in the design?

A51. A

Q52. For articulation-point queries between a specified source and target, should the source and target themselves be excluded from the reported articulation points?

A52. Yes.

Q53. If no path originally exists between the source and target, should the result simply report that no articulation point exists?

A53. Yes.

Q54. Scope of Low-Level Rewiring Prompts Should hidden testcases be expected to contain direct low-level rewiring instructions that may change circuit functionality?

For example:

Reconnect input pin A of gate g1 from net n1 to net n2. or

Replace the driver of net n5 with gate g3. If such an operation changes the circuit function, should the system:

- A. Execute the rewiring literally because it was explicitly requested; or
- B. Apply it only when functional equivalence with the original design can be proven?

Alternatively, should we assume that modification prompts will primarily request equivalence-preserving transformations, such as cleanup, constant propagation, technology mapping, buffer insertion, or optimization?

A54. All the modification prompts could assume no functional change. All the equivalence should be preserved.

Q55. Is the use of a packaged Miniconda environment allowed?

To resolve dependency issues, we are considering packaging our entire Miniconda environment (e.g., using conda-pack) and submitting it along with our executable. Our entry script would then unpack and activate this local conda environment to run our Python agent. Could you confirm if this deployment method is permitted and technically feasible on the TSRI machine?

A55. During evaluation, we execute the program with `./cada<team_number>_<beta|final> -config <config_file_path>`. Make sure your conda environment can run with this command.

Q56. Are contestants allowed to use third-party EDA tools, such as OpenROAD or an LEC tool, after packaging them as executable binaries together with the submission?

A56. Yes.

Q57. Are we allowed to include the required shared libraries, scripts, configuration files, and other dependencies needed by these binaries?

A57. Yes.

Q58. Are contestants allowed to invoke a self-provided Podman or Docker executable from the submitted program?

A58. No.

Q59. Are there any restrictions on nested containers, rootless execution, namespaces, mounts, extended attributes, or network access that we should account for?

A59. During evaluation, we will copy the contestant submission to our evaluation path on the TSRI machine. Please make sure your program can be executed with `./cada<team_number>_<beta|final> -config <config_file_path>`.

Q60. We got NOT_FOUND in Alpha Test. But as we tested on the submission TSRI machine, our program indeed output a log file under the testcase folder, the directory is:

```
└── testcase
    └── test01
        └── test01.log
```

Does this directory fit the demand of the problem? Besides, will there be "testXX" folders between the "testcase" folder and .v file during the test?

We also install a tool relied on by our program on the submission TSRI machine. We modified the environment variable, and our program uses bash commands to run the tool.

So we wonder whether the Alpha Test was run on the submission TSRI machine or our program was copied to another environment for evaluation?

A60. Please put the log file and output netlist under the executable program path or the path to the given input design path within the input prompt. We will copy the executable to different path, but still on the TSRI machine.

e.g.,

```
└── cada1000_alpha
    └── test01.log
```

Q61. If an earlier request performs optimization that may legally remove or rename internal gates/wires, and a later request refers to an original identifier such as g0 or n74, Should the system preserve original identifiers/aliases across transformations, or should later references be interpreted only in the current transformed netlist?

A61. In the current transformed netlist.

Q62. For prompts phrased as ‘try to reconnect/replace/optimize X while preserving functionality,’ if the requested edit cannot be applied without changing functionality, is a verified no-op response acceptable for full credit, provided the system clearly states that the original implementation was retained?

A62. Yes.

Q63. When multiple transformation requests occur in one testcase, should structural constraints from earlier requests remain hard constraints for the final netlist after later transformations? For example, if request 1 enforces max fanout 4 and request 2 remaps the entire design, must the final design still satisfy max fanout 4?

A63. Yes.

Q64. If a prompt asks to optimize after buffer insertion or basis remapping, is the cost measured on the final constrained netlist including inserted buffers/remap gates, or only on the optimized logic before enforcing the constraint?

A64. The cost is evaluated on the final netlist.

Q65. For a prompt asking to optimize or analyze the fanin cone of a signal that is a DFF.Q/register output, should the cone be treated as empty/combinational boundary per Q21, or should it refer to the D-input combinational logic of that same flip-flop?

A65. Treat it as the boundary per Q21.

Q66. If a request says ‘replace all X gates in the cone of Y’ and the current cone contains no such X gates, is reporting zero replacements with equivalence preserved considered fully correct?

A66. Yes.

Q67. When a prompt says the final netlist must use only a specified gate basis, are temporary helper gates of other types allowed during intermediate transformations as long as the final written netlist satisfies the basis?

A67. Yes.

Q68. For transformation requests, does the judge score only the generated Verilog and final metrics, or can the natural-language response affect transform credit if the Verilog is correct, but the summary is brief?

A68. We verify the transformation prompt using only the generated Verilog file.

Q69. We have three clarification questions on the Q&A semantics, each with a concrete example from the released testcases.

1. DFF enable/hold recognition (Q46/Q47): should EN and DATA be existing nets in the netlist, or is any Q-free Boolean decomposition acceptable (i.e., the check is purely that the D-input function is positive-unate in Q with true Q-dependence)? For example, in public test40 the flip-flop g1274 has $D = (Q \& n1850) \mid (n977 \& n7677)$; the two guard terms are not complementary nets, but a function-level reading finds valid EN/DATA functions. Which reading should be used when counting enable/hold structures?

2. Register-to-register path counting (e.g., public test32 and test37): should the count be the number of distinct combinational paths, or the number of distinct (source DFF, sink DFF) pairs, and should self-loops (a DFF's Q feeding back to its own D) be included? Published team answers diverge by orders of magnitude on exactly this definition, so a one-line ruling would settle it.

3. Floating primary outputs in transformation requests: several released netlists declare primary outputs with no driver (e.g., public test25 declares output [3:0] n13 which is never driven), consistent with Q&A A5.6. For transformation requests, should the output netlist also leave such ports undriven, or is tying them to a constant acceptable? Equivalently, is the official equivalence check sensitive to a previously-floating output acquiring a constant driver? We currently preserve the floating form to

be safe under either interpretation, and would simply like to confirm this is the expected behavior.

A69.

1. A Q-free Boolean decomposition is acceptable. The example you mentioned is a valid EN/DATA function.
2. Preserve the floating form. Undriven primary outputs must remain undriven in the output netlist.
3. Preserve the floating form. Undriven primary outputs must remain undriven in the output netlist.

Q70. Many of our test cases failed with log file not found or similar path-related errors. If a test case's input file is located at `workdir/testcase/case_name/input.v`, should our agent generate the `<case_name>.log` and the output Verilog files in that exact same directory (`workdir/testcase/case_name/`), or should they be generated in the root `workdir/` directory?

A70. Generating the log under either `workdir/` or `workdir/testcase/case_name/` is acceptable. Please note that the testcase folder name might change. For example, it could be `workdir/testcase_alpha/case_name/`. We will specify the design path in the prompt.

Q71. We would like to request clarification regarding the evaluation environment and the expected locations of generated files.

The specification states that the executable will be invoked as:

```
./cada0001_alpha -config <config_file_path>
```

Could you please clarify the following?

1. From which working directory will the executable be invoked? Will it be the root directory of the submitted package or a testcase-specific directory?
2. Where should the `<case_name>.log` file be created? The testcase request provides only a filename, such as `case23.log`, without a full path.

3. Similarly, when an output netlist filename such as case28_out.v is provided without a directory, should it be written to the executable's current working directory?
4. Will a separate working directory be provided for each testcase, or will the outputs of all testcases be written to a common directory?

A71.

1. We will execute the program in the working directory (e.g., alpha_test_submission) for each team.
2. See Q60.
3. Yes.
4. A common directory (the path to the executable program) will be used.

Q72. A65: does the boundary reading also apply to transformations?

A65 says that when the target of a fanin-cone request is itself a DFF.Q output, the cone is treated as the combinational boundary.

Under that reading the cone is empty, so for a request such as "Replace all 2-input OR gates in the cone of n11[0] with equivalent logic built only from NAND and NOT gates" (released 0813 testcase test25, request 4 ; n11[0] = a flip-flop Q) the correct behaviour would be to replace nothing and report zero replacements (which A66 and A62 appear to permit). The alternative reading is to operate on the D-input logic of that same flip-flop, which is what our Beta submission did. Which behaviour is scored as correct — for transformation requests, and for analysis requests such as "report the number of each gate type in the cone of X" or "which output has the largest fanin cone"

A72. For the transformation prompt we consider both correct. As for the analysis prompt, the DFF should be the fan-in cone boundary.

Q73. Q69 asked three questions. In A69, items 2 and 3 give the same answer about floating primary outputs, and Q69.2 — register-to-register path counting — does not appear to have been answered.

Could you confirm whether a register-to-register "path" is counted as (a) the number of distinct combinational paths, or (b) the number of distinct (source flip-flop, sink flip-flop) pairs, and whether self-loops (a flip-flop's Q feeding back to its own D)

count? For released 0813 testcase test58 the two readings give 107 and 36 respectively, so the choice changes the answer by a factor of three.

A73. Count the distinct combinational paths from any DFF.Q to any DFF.D, with self-loops included.

Q74. For the prompt:

“Report the number of each gate type in the cone of the signal X.”

we would like to clarify the expected behavior when signal X is the Q pin of a D-Flip-Flop.

Should the D-Flip-Flop itself be included as a gate in the fanin cone of X?

Alternatively, should we follow the same convention as Q21 and treat the Q pin of a D-Flip-Flop as a pseudo primary input (pseudo PI), in which case its fanin cone would be empty?

A74. DFF itself is included.

Q75. For the prompt:

“List all register-to-register paths in this design through combinational logic.”

we would like to clarify the definition of a register-to-register path.

Should we consider all distinct paths between every pair of registers, including paths from a register back to itself (self-loops), as long as the path goes through combinational logic?

If so, for some designs the number of such paths can be very large. In that case, would it be acceptable to report only the total number of register-to-register paths and provide several representative paths as examples, rather than listing every path explicitly?

A75. Count all distinct combinational paths from DFF.Q to DFF.D, including self-loops, and provide complete enumeration. We have also updated the test cases which do not contain the issue you mentioned.

Q76. If a prompt first asks us to rename a gate or signal, and a subsequent instruction asks us to optimize the circuit (for example, reducing the logic depth), is the optimized circuit required to preserve the previously renamed gate or signal?

In other words, should the renamed object be treated as a constraint that must remain in the final optimized netlist, even if the optimization could otherwise remove or restructure it?

A76. If a later prompt overrides an earlier rename, you do not need to preserve the renamed gate/signal in the final optimized netlist.

Q77. The result sheet has three columns — Basic, Transform, Opt — while Section 4 defines Analysis (§4.2) as a separate task family. Could you confirm whether analysis questions are scored under Basic, or whether they are not scored in this sheet?

This also affects the runtime limit: Section 3.1 and A18 allow 60 seconds for basic operations and 300 seconds for all other requests. If analysis questions are counted as Basic, would the 60-second limit apply to them?

A77. Analysis questions are counted as Basic.

Q78. Our Beta Test evaluation results showed NO_LOG for every test case.

We would like to clarify whether the prompt or input format used to provide the test case name during evaluation may differ from the example test cases, or whether it remains the same.

This information would help us determine whether the issue is related to how our program handles the test case name and generates the log file, or whether there may be another underlying issue with our submission.

A78. We will copy the testcase folder to the workdir and modify the testcase path in the prompt to an absolute path (e.g., load the design from the file test01.v located in the directory /path/to/workdir/testcase_beta/test01/).

Q79. Regarding the number of Reg-to-Reg paths in the public test cases, should we report the number of Reg-to-Reg paths or the number of Reg-to-Reg pairs?

A79. Reg-to-Reg paths.

Q80. A65: does the boundary reading also apply to transformations?

A65 says that when the target of a fanin-cone request is itself a DFF.Q output, the cone is treated as the combinational boundary.

Under that reading the cone is empty, so for a request such as "Replace all 2-input OR gates in the cone of n11[0] with equivalent logic built only from NAND and NOT gates" (released 0813 testcase test25, request 4 ; n11[0] = a flip-flop Q) the correct behaviour would be to replace nothing and report zero replacements (which A66 and A62 appear to permit). The alternative reading is to operate on the D-input logic of that same flip-flop, which is what our Beta submission did. Which behaviour is scored as correct — for transformation requests, and for analysis requests such as "report the number of each gate type in the cone of X" or "which output has the largest fanin cone"

A80. For the transformation prompt we consider both correct. As for the analysis prompt, the DFF should be the fan-in cone boundary.

Q81. A69 appears to have a numbering mismatch

Q69 asked three questions. In A69, items 2 and 3 give the same answer about floating primary outputs, and Q69.2 — register-to-register path counting — does not appear to have been answered.

Could you confirm whether a register-to-register "path" is counted as (a) the number of distinct combinational paths, or (b) the number of distinct (source flip-flop, sink flip-flop) pairs, and whether self-loops (a flip-flop's Q feeding back to its own D) count? For released 0813 testcase test58 the two readings give 107 and 36 respectively, so the choice changes the answer by a factor of three.

A81. Count the distinct combinational paths from any DFF.Q to any DFF.D, with self-loops included.

Q82. Could you please clarify the exact contract expected by the judge?

1. Does the judge follow the file path included in the natural-language response and read the referenced file when evaluating complete-enumeration queries?
2. Path form: If so, should the path printed in the answer be relative to a specific base directory (for example, the Docker working directory /app or the executable directory), or is an absolute path also acceptable? If a relative path is required, what directory is it resolved against?

3. File location: Is a per-testcase <case_name>_query_results/ subdirectory acceptable, or must the dumped file be placed directly under the executable path or input-design path, as we understand from Q60?
4. File format: Does the dumped file need to contain a natural-language answer, or is a structured format such as JSON acceptable as long as it contains the complete requested result?
5. Truncated terminal responses: If the terminal response is truncated (for example, it contains only a capped preview plus a reference to the dumped file), does this cause the grader to ignore the referenced file and evaluate only the truncated terminal/dialogue content?

A82.

A1-A2: Yes, the judge follows file paths and reads referenced files. We have updated the scores.

A3: Per-testcase subdirectories are acceptable.

A4: Both natural-language and JSON formats are acceptable.

A5: No, we evaluate the referenced file even if the terminal response is truncated.

Q83. We would like to clarify the expected handling of MUX-related structures and enable/hold analysis in the hidden test cases.

In a previous Q&A, it was stated that the input Verilog would follow Section 3.2 and use only the following supported primitive gates:

AND, OR, NAND, NOR, NOT, BUF, XOR, XNOR, and DFF.

In the latest reply, it was mentioned that we should handle structures that have not appeared in the released benchmarks, including possible MUX-related structures.

We would like to clarify how these statements should be interpreted.

Hidden netlist syntax

Will hidden netlists still use only the primitive gate types listed above, while the prompts may refer to higher-level MUX functionality implemented using combinations of those gates?

Or may hidden netlists directly contain a native MUX primitive or another previously unspecified gate or cell type?

If native MUX primitives may appear, could you provide the exact Verilog keyword, pin order, and connection syntax that will be used?

A83. Hidden netlists still use only the primitive gate types.

Q84. Functionally equivalent MUX implementations

If hidden netlists still use only the original primitive gates, should we recognize any gate-level implementation that is functionally equivalent to a MUX, even when it does not match a canonical AND/OR, NAND, or NOR MUX structure?

For example, consider:

$$D = Q \text{ XOR } (EN \text{ AND } (Q \text{ XOR } DATA))$$

This function is equivalent to:

$$D = EN ? DATA : Q$$

Should this be recognized as a MUX-based enable/hold structure, even though its structural form is different from a canonical MUX implementation?

A84. Yes.

Q85. AND-gated D-input structures

For a DFF whose D input is implemented as:

$$D = EN \text{ AND } DATA$$

there is no Q-feedback path. When EN is inactive, D is forced to 0 rather than holding the previous Q value.

Should this still be reported as an enable/hold structure, or should it only be classified as a gated-data pattern?

More specifically, must a structure satisfy $D = Q$ under the inactive enable condition in order to be counted as a hold structure?

A85. No, this should not be considered.

Q86. Multiple or nested patterns

If the same D-input function has multiple valid enable/hold decompositions, should all valid decompositions be reported, or is one functionally proven decomposition sufficient?

If MUX patterns are nested or overlapping, should we report only the outermost control structure or every valid nested pattern?

For count-related prompts, should the result count unique DFFs with a recognized pattern, or the total number of recognized patterns?

A86. One functionally proven decomposition is sufficient.

Q87. For a gate-level circuit, when we are asked to find articulation points, should an articulation point refer to:

a gate only,
a net only, or
either a gate or a net?

A87. A gate only, not a net.

Q88. If we are asked to count the number of NOT gates, should a NOR gate be counted as 1 NOT gate plus 1 OR gate, or as 0 NOT gates?

A88. 0 NOT gates; NOR is its own primitive type and is not decomposed.

Q89. Q18 states that Basic requests have a 60-second time limit and other requests have a 300-second time limit. Q77 later states that Analysis questions are counted as Basic.

Could you please confirm the intended runtime rule?

1. Are all Analysis requests subject to the 60-second Basic-request time limit?
2. Does that limit include complete result generation, writing and closing any spill/result file, log serialization, printing and flushing the final `#END <id>`, and all provider/tool calls for that request?
3. For an Analysis request that explicitly requires complete enumeration of a very large result, such as all register-to-register paths under Q75/Q79, is the same 60-second limit still applied?
4. If the request reaches the time limit, is an incomplete count/sample considered incorrect even when the full result cannot be published in time?

A89. Yes, the 60-second Basic limit applies to all Analysis requests, including full result/file/log finalization, and an incomplete result at timeout is scored as incorrect.

Q90. The Problem A specification states that a comparable LLM service may be substituted if a designated service has an outage or abnormal behavior, while Q&A

A6 states that different models would be used only if both named models are deprecated.

For implementation compatibility, could you please clarify the contestant-facing interface if any replacement is made? Specifically, will the evaluation configuration continue to use `provider: openai` or `provider: anthropic` with the same corresponding API/config schema, possibly with only the model or endpoint changed?

A90. Yes, the same provider: openai/anthropic config schema will be used, with only the model or endpoint changed if needed.

Q91. Q&A A16, A21(3), and A75 require a complete large enumeration to be written to a file and the response to provide the file path. For Final compatibility, when that result file is created in the executable's working directory, may the response provide a path relative to that working directory, or must the returned path be absolute?

A91. Both are acceptable.

Q92. The initial netlist may not have dangling items directly, but it may have some after a re-synthesis process of some tools.

So, if we are asked to remove dangling items, should we consider those appear after re-synthesis processes?

A92. If the prompt asks to remove the dangling items on the initial netlist without dangling items, you do not need to remove anything. If the prompt asks to remove the dangling items after any transformation, you have to remove the dangling items in the transformed netlist.

Q93. We would like to follow up on Q77 regarding the classification of analysis questions.

In A77, it was clarified that analysis questions are counted as Basic. We would like to further confirm whether this classification also applies to the runtime limit.

According to Section 3.1 and A18, basic operations have a 60-second runtime limit, while all other requests have a 300-second limit. Therefore, should analysis questions also use the 60-second runtime limit?

A93. Yes.

Q94. We have a follow-up question regarding A74, on whether bounding DFFs should be counted in a fan-in cone:

1. A74 clarifies that for the prompt "Report the number of each gate type in the cone of the signal X", when X is the Q pin of a D-Flip-Flop, the DFF itself is included in the cone.

We would like to confirm the expected behavior for the more general case: when X is a combinational signal whose fan-in cone terminates at the Q pins of one or more D-Flip-Flops (treated as pseudo primary inputs per Q21), should those bounding DFFs also be counted as gates in the cone of X?

2. For example, consider and g1 (X, q0, q1); where q0 and q1 are the Q outputs of flip-flops g2 and g3. Should the cone of X be reported as 1 gate (AND: 1 only, the DFFs are boundaries and excluded), or as 3 gates (AND: 1, DFF: 2, the bounding DFFs are included in the same spirit as A74)?

3. Additionally, does the same convention apply to the prompt "How many gates are in the fanin cone of X" (the plain gate-count wording)? In particular, when X itself is a DFF.Q output, should the answer be 1 (the DFF itself, consistent with A74), or 0 (empty cone per Q21/A65)?

A94.

1. Yes — bounding DFFs are counted.
2. 3 gates (AND:1, DFF:2).
3. Yes, same convention — answer is 1, not 0.